

Novel eSIM Forensic Framework for eSIM-Enabled M2M and IoT Devices

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Abstract—The rapid adoption of IoT and M2M devices has brought unparalleled connectivity to diverse industries. At the core of many of these devices is eSIM technology, a remotely manageable and embedded solution that has revolutionized connectivity by eliminating the need for physical SIM cards. Modern IoT and M2M devices, such as Vehicle Tracking Devices (VTD), On-Board Diagnostics (OBD) and Telematic Devices, feature eSIM profiles that store forensic-relevant information crucial for digital forensic investigations. We proposed and developed the Novel eSIM forensic framework as a proof-of-concept for conducting forensic analysis on eSIM-enabled IoT and M2M devices. By examining the procedural and practical aspects of forensic data extraction from eSIM-enabled M2M devices, we provide valuable insights and methodologies for the forensic community. This research underscores the importance of eSIM forensic analysis, equipping investigators with cutting-edge tools and insights to navigate complex IoT and M2M ecosystems.

Index Terms—IoT Forensics, M2M, eSIM, Digital Forensics, Reverse Engineering, Device Forensics, Vehicle Tracking Device

I. INTRODUCTION

The Internet of Things (IoT) and Machine-to-Machine (M2M) technologies are revolutionizing industries by enabling interconnected devices to communicate and share data. In healthcare, they improve patient monitoring and management. In manufacturing, they enhance automation and predictive maintenance. In transportation, they optimize fleet management and logistics.

New IoT and M2M technologies are coming with different internet connectivity options; one of the modern solutions for the internet connectivity is embedded SIM (eSIM) for M2M. An electronic device with a cellular connection has an embedded Universal Integrated Circuit Card (eUICC) on its circuit board called eSIM, which helps to establish cellular connectivity on specific devices.

The GSM Association (GSMA) has defined two common eSIM use cases [1] the first, known as M2M eSIM for IoT devices, and the second is for end-user consumer devices. M2M communication is a type of data communication in which one or more entities are involved, but human involvement or intervention is not always necessary for the communication process to occur. This technology allows billions of devices to connect over the internet, giving rise to a number of new applications, services, and business opportunities at various levels.

As these technologies make rapid strides in all areas of society, they create unique challenges and opportunities in

IoT forensics. When number of interconnected devices grows, the complexity of investigating security breaches and data incidents within these networks also set to grow.

IoT forensics involves the meticulous process of analyzing data from a wide range of smart devices to uncover digital evidence. As the complexity and volume of IoT systems continue to expand, the demand for sophisticated forensic techniques becomes essential, and our research work highlights the need for forensic advancements in eSIM-integrated IoT/M2M technologies. Our objective is to collect and analyze evidence and artifacts from the IoT/M2M ecosystem using advanced forensic techniques.

Forensic challenges of eSIM include secure data extraction due to its embedded nature, encryption complexities, and limited accessibility. Investigators must navigate proprietary technologies, varied eSIM implementations, and the necessity for specialized tools. Ensuring data integrity while maintaining chain-of-custody protocols also poses significant hurdles. Therefore, it's crucial to conduct research and develop an effective forensics framework for eSIM-integrated IoT technologies. In order to develop such a framework, we conducted a real-time forensics approach in Vehicle Tracking Device (VTD) at the Digital Forensics Lab, C-DAC. The contribution of this research is as follows:

- Case Study on Integrated GSM/GPRS Modules with eSIM in Vehicle Tracking Devices (VTDs).
- Innovative Forensic Framework for eSIM in IoT and M2M Applications.
- Digital Forensic Challenges (DFI) in eSIM-Integrated IoT and M2M Technologies.

Our research aims to deliver a robust and scalable digital forensic solution for eSIM-integrated IoT/M2M devices, facilitating improved investigative capabilities in this rapidly advancing field.

II. RELATED WORK

This section reviews existing research on eSIM technology and its applications in the IoT and M2M domains. The studies primarily focus on digital forensic approaches to eSIM and the associated challenges in conducting investigations within IoT and M2M ecosystems.

Alshenaifi et al. [3] proposed a framework-based forensic solution for eSIM-integrated M2M devices along with challenges and issues. However, they did not mention the

practicality of performing digital forensics analysis in a real-time scenario. And most of the research work is about how to carry out an investigation in M2M eSIM technology. From the suggested framework model, if the investigator goes with chip off, there is a possibility that the system/device state data might get lost due to direct contact with the circuit board. And if the investigator plans to go with the investigation by getting the credential from the suspect, this takes a good amount of time, and the investigation gets further delayed. Our research work overcomes these challenges by implementing a systematic investigation strategy.

Zawoad et al. [4] proposed a Forensics-aware IoT (FAIoT) model, which relies on a centralized trusted evidence collection repository located in the cloud. In this model, investigators collect evidence from the respective repository and conducts analysis for further investigation. The proposed model includes a secure logging scheme to ensure the reliability of the evidence. The proper functioning of this model depends on the number of IoT devices registered with the secure evidence repository services. If a device is not registered with the secure evidence repository service, collecting evidence becomes challenging. Our framework overcomes this challenge by implementing device-level forensics in IoT/M2M devices.

Amirhosein et al. [5] conducted a study on various security challenges and attacks in M2M communications. Their research addresses the security requirements for M2M communication, along with the types of vulnerabilities and challenges. The study also explores attack scenarios in M2M, including software, hardware, and network-level attacks. This theoretical study clearly indicates the need for cyber forensic solutions for hardware-level attacks. We directly address hardware-level forensics through the novel eSIM framework.

M2M technology opens up a vast array of new applications, services, and commercial prospects across multiple industries by enabling billions of devices to connect over the internet. M2M technology also transforms the performance of numerous verticals of diverse industries, enterprises, and services. The different use cases of the eSIM and iSIM, categorized into 30 application groups, are mentioned in this white paper [6]. The eSIM technology picks up its momentum in both the consumer and M2M ecosystems. Thus the demand for conducting digital forensics investigations becomes crucial in incidents involving such devices, and our framework provides valuable insights and methodologies for the forensic community.

III. RESEARCH PROBLEM

As eSIM technology advances, it also creates new security challenges and concerns in IoT forensics. It raises the need to conduct more research and develop effective forensics frameworks in eSIM-integrated IoT/M2M technologies. Currently, there is no practical research on eSIM forensics for IoT/M2M devices, except a framework model [3]. Our main objective is to research, develop, and implement an effective framework for eSIM forensic analysis in IoT and M2M devices. This includes conducting real-time experiments and analyses, such as forensic analysis of integrated GSM/GPRS modules with

eSIM technology in Vehicle Tracking Devices (VTD). By implementing our framework, we addressed existing challenges in conducting Digital Forensics Investigation (DFI) on eSIM-associated IoT/M2M technology.

IV. FORENSICS APPROACH TOWARDS TELEMATIC DEVICES

This section provides an overview of the types of telematic devices used for vehicle tracking and performance monitoring applications.

A. VTD and OBD Vehicle Devices

Vehicle Tracking Device (VTD) is used for monitoring the location and movement of a vehicle, and On-Board Diagnostics (OBD) is a computerized system within a vehicle that monitors and controls the vehicle's performance. These devices are integrated with eSIM technology for telematic efficiency.

B. VTD and OBD with Cloud Ecosystem

A vehicle tracking system enables the monitoring and management of vehicles with the help of a computer, smartphone, or tablet. The VTD/OBD with cloud ecosystem is given in Figure 1. These systems facilitate real-time and historical tracking of vehicle speeds, travelled routes, stop points, and idling durations, all displayed on maps. These data are updated 24/7 to a central server that includes both historical and current reports.

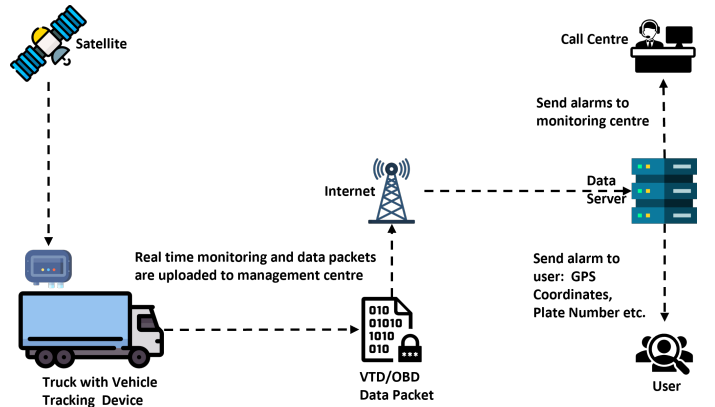


Fig. 1. VTD with cloud ecosystem

V. SYSTEM ARCHITECTURE

This section details our system architecture, which integrates the eSIM/eUICC ecosystem for M2M devices, the Quectel M66 chipset, and the usage of the AT Commands library. For eSIMs, we follow the standards recommended by the international standard for eSIM published by the GSM Association [1] and the European Telecommunications Standards Institute [2]. This section is organized as follows:

- eSIM/eUICC Ecosystem for M2M
- Quectel M66 Chipset for Telematics
- AT Commands Library
- Sensitive Data in eSIM

A. eSIM/eUICC Ecosystem for M2M

There are two groups of GSMA eSIM specifications currently exist: consumer solution architecture and machine-to-machine (M2M) solution. eSIM terminology includes following components:

- **eSIM:** The SIM hardware that is permanently embedded in to a device.
- **eSIM profile:** The eSIM profile holds user data, mobile operator credentials and data related to a subscription. Profiles are remotely downloaded to the eUICC of the device.
- **eUICC or embedded Universal Integrated Circuit Card:** A term used to describe an Operating System (OS) and software, combined with eSIM hardware, which enables a device to be configured and managed via Remote SIM Provisioning (RSP).

The consumer eSIM solution and the M2M solution use distinct but comparable architectures. Both use common features such as RSP system and secure channels. The key differences are the direction of control and server management relationship. The consumer solution relies on local end-user control: consumers trigger new profile downloads as they select operators through their device interfaces. In M2M, profiles are managed remotely, usually by a third party, without any further human intervention. The eSIM ecosystem for M2M/IoT is given in Figure 2. The M2M eUICC can be managed only by a single, pre-determined M2M RSP server as opposed to the consumer eUICC, which can communicate with and receive profiles from any compliant consumer RSP server [1].

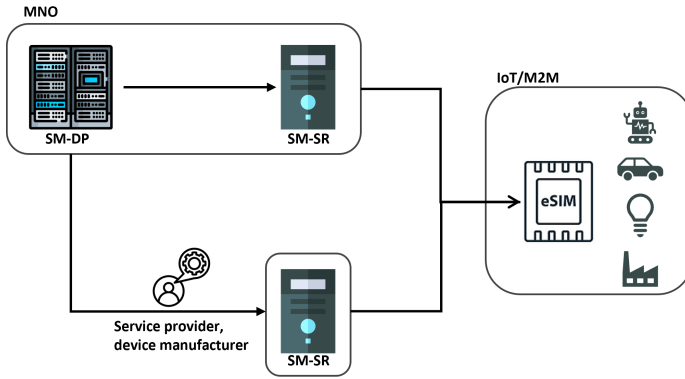


Fig. 2. eSIM ecosystem for M2M

B. Quectel M66 Chipset for Telematics

The Quectel M66 is the tiniest quad-band GSM/GPRS module [7]. The M66 chipset comes with $15.8 \times 17.7 \times 2.3$ mm dimensions, making it an ideal platform for applications that are sensitive to size. M66 can be readily integrated into low-volume applications and offers dependable connectivity because of its compact LCC package size and low profile. The module can serve a range of applications in wearables, automobiles, industrial PDAs, smart metering, telematics, etc.

C. AT Commands Library

Single-line commands known as AT commands can be transmitted to the device in a specific order, and the user must wait for a response before delivering another command. We use a USB Serial UART (Universal Asynchronous Receiver/Transmitter) converter to transmit the relevant AT commands to the VTD module and wait for its response. AT commands are of 4 types based on their usage: test, read, set and execution commands. For testing purposes, we used only test and read commands [8]. For the AT commands library interface, the user can interact with a connected serial bridge (USB device) with the help of either of these software: using PuTTY (serial communication via terminal) or m-center (serial communication via GUI) from u-blox.

D. Sensitive Data in eSIM

Forensic data collected from a traditional SIM profile includes Integrated Circuit Card Identifier (ICCID), International Mobile Subscriber Identity (IMSI), Service Provider Name (SPN), Mobile Country Code (MCC), and Mobile Network Code (MNC) [9]. The file structure of a normal SIM module includes Main Files (MF), Dedicated Files (DF), and Elementary Files (EF) [10].

Similarly, eSIM profile also contains a significant amount of sensitive information about the subscriber, vendor, and device, which holds forensic value. In this context, we gather both eSIM-specific and device-specific data for Forensic Data Analysis (FDA).

VI. SYSTEM IMPLEMENTATION

For implementation and analysis purposes, we are using VTD or a telematics device integrated with a GSM/GPRS module with eSIM.

A. System Design and Setup

For extracting forensically relevant data from the VTD (which comes with an eSIM and Quectel-M66 GSM/GPRS chipset), we use a serial I/O read and write interface (e.g., RS232/422/485, I2C, USB, etc.), which acts as an interface between the Quectel M66 module and the system. After that we use a set of proprietary AT commands to retrieve and manipulate relevant data from the eSIM and VTD.

B. System Implementation Stages

In each stage we ensured that the implementation stage meets all the criteria and requirements. All activities are carried out in a well-controlled and monitored environment. The overall system design is given in Figure 4.

1) *Implementation Stage 1 - Workstation Setup:* Verify that the computer meets the hardware and software requirements. Both Linux and Windows environments support its deployment. Testing activities are carried out in a controlled lab environment with field experts.

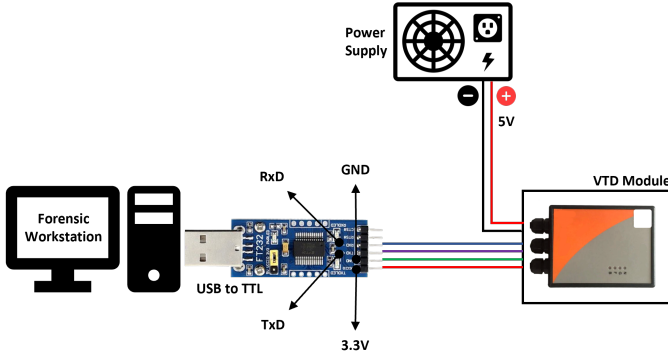


Fig. 3. VTD forensics data analysis setup

2) *Implementation Stage 2 - Driver Module Setup:* The FDA workstation setup includes software and driver setup. The CP2102 USB-to-UART Bridge Controller is used to provide USB connectivity while communicating via a simple serial protocol. It is the key component of our setup (TTL to USB adapter). Along with the converter, we used the CP2102 USB-to-Serial Bridge Driver. The user can interact with the connected serial bridge (USB device) using either PuTTY or m-center.

3) *Implementation Stage 3 - Circuit Setup:* In VTDs, pin mapping will vary based on different manufacturers. It is mandatory to refer to the datasheet of the eSIM and M66 GSM/GPRS module chipset [7]. Pin selection is based on the M66 GSM/GPRS datasheet, and verified the Transmit Data (TxD) and Receive Data (Rx) pins for continuity. To enable data communication, we implemented a connection between the TTL to USB (CP2102) connector and the M66 GSM/GPRS module. If somehow data communication via Pin Selection-A fails, then investigator can go with Pin Selection-B. The pin selection between M66 and TTL to USB is given in Table 1.

TABLE I
PIN SELECTION: M66 AND TTL TO USB CONNECTOR

M66 Pin Selection-A	M66 Pin Selection-B	TTL to USB
Pin - 17 - TxD	Pin - 39 - DBG TxD	RxD
Pin - 18 - Rx	Pin - 38 - DBG Rx	TxD
Pin - 42/43 - VBAT	Pin - 42/43 - VBAT	3V3
Pin - 41 - GND	Pin - 41 - GND	GND

4) *Implementation Stage 4 - Evidence collection:* Set up circuit hardware for the VTD device and proceed with forensic evidence collection and analysis of the eSIM-integrated GSM/GPRS module.

VII. FORENSIC DATA ANALYSIS (FDA)

Our primary focus was on performing FDA and identifying data with forensic relevance that can be extracted during the process. These data are collected using m-center or monitored through a serial connection using PuTTY. The overall process is shown in Figure 4.

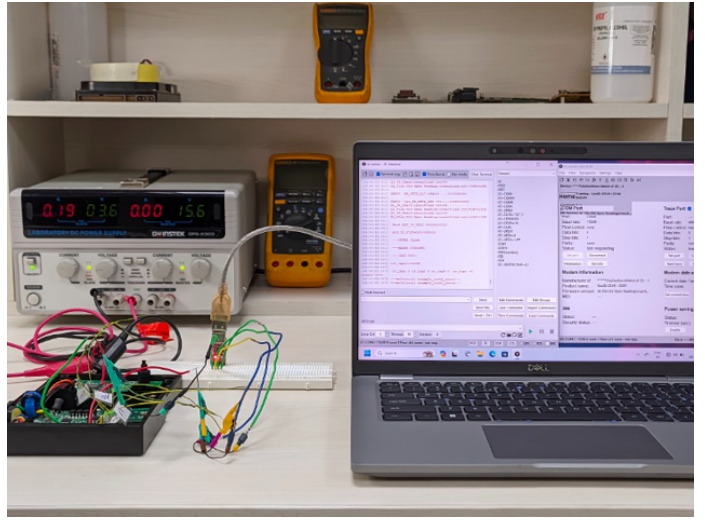


Fig. 4. VTD forensics data acquisition

A. FDA: Integrated GSM/GPRS Module with eSIM

The sensitive data in VTD eSIM includes the following: International Mobile Equipment Identity (IMEI), Integrated Circuit Card Identification (ICCID), Internet Protocol (IP) address and Port Number, Web Server URL, LOC (Location), RTO Number (Vehicle Registration Number), Emergency Response Number, etc. A few of the mentioned details that are collected through the framework are shown in Figures 5, 6, and 7. These results provide valuable insights into eSIM forensics analysis in IoT/M2M-connected devices.

```
[14:28:30.186] rx: --startup at cmd---
[14:28:30.186] rx: IMEI_NUMBER=868996037
[14:28:30.186] rx:
[14:28:30.186] rx: CCID_NUMBER=8991047312180
[14:28:30.798] rx: <--SIM card status is normal! simStat 1-->
[14:28:31.611] rx: <-- 2 Network State:creg=2,cgreg=2 w_sw_cnt=0-->
[14:28:31.689] rx: GeoID: 22341 : 22341
GEOFID_Ret:22341

GeoAlert: 2 : 2
GeoAlertTypeRet:2
```

Fig. 5. FDA analysis result (1)

```
[17:19:43.922] rx: FIRMWARE VERSION : 05M66_FVERSION_A0T_AIS140_22_10_2019_09.16pml89*
[17:19:43.922] rx: 05M66_FVERSION_AIS140_22_10_2019_09.16pml89*
[17:19:43.922] rx: IGNI SLEEP , FIX IS MUST
[17:19:43.922] rx: Free RAM=647668 / 647668
[17:19:43.922] rx: Free Flash=281088 / 301568
[17:19:43.922] rx: IIP Address118.067.240.086
[17:19:43.922] rx: Port=02021
[17:19:43.922] rx: SOC1=1
[17:19:43.922] rx: Update_Timer=020
[17:19:43.922] rx: APN=sensem2m
[17:19:43.922] rx: Owen_number=1112223334
[17:19:43.922] rx: url_port_v=02021
[17:19:43.922] rx: url_v=mda3.fleetvigil.co.in
[17:19:43.937] rx:
url_port_e=02021
[17:19:43.937] rx: url_e=mda2.fleetvigil.com
[17:19:43.937] rx: HTTP URL =http://13.229.232.186:80/
[17:19:43.937] rx:
[17:19:43.937] rx: fresult 0.000000
[17:19:43.937] rx:
[17:19:43.937] rx: OLD_odo_COUNT=000000
```

Fig. 6. FDA analysis result (2)


```

[17:19:45.921] rx: RTO =^^,000000KL01AX
[17:19:45.921] rx: or_str_rd_snd_num=000000
[17:19:45.921] rx: or_str_rd_snd_num=0
[17:19:45.921] rx: Ql_FS_Read():=0: readedlen=11
[17:19:45.937] rx: strBuf=1234567890
[17:19:45.937] rx:
[17:19:45.937] rx: Fallback Sent_Message_number=
[17:19:45.937] rx: LAST VALID LOC=008.509937,N,076.961910,E
[17:19:45.937] rx: 008.509937,N,076.961910,E
[17:19:45.937] rx:
[17:19:45.937] rx: kr_lat 008.509937 kr_lon 076.961910
[17:19:45.937] rx:
[17:19:45.937] rx: EMERGENCY R
[17:19:45.953] rx: eply_number=
[17:19:45.953] rx: Ql_FS_Read():=0: readedlen=4

```

Fig. 7. FDA analysis result (3)

B. Digital Forensics Investigation: Location Tracing

Investigator can use insights received from the framework for further investigation. One such instance is tracing and locating the data server, which holds GPS historical data. With this, the investigator can trace the exact location (LOC) related to the incident in question. We were able to trace the exact location successfully using GPS coordinates obtained during our experiment to **kr_lat 8.509937 kr_lon 76.961910**. The LOC tracing result is shown in Figure 8.

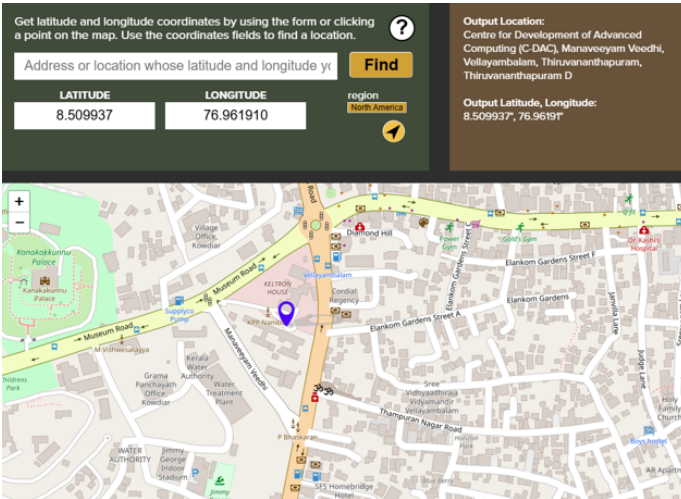
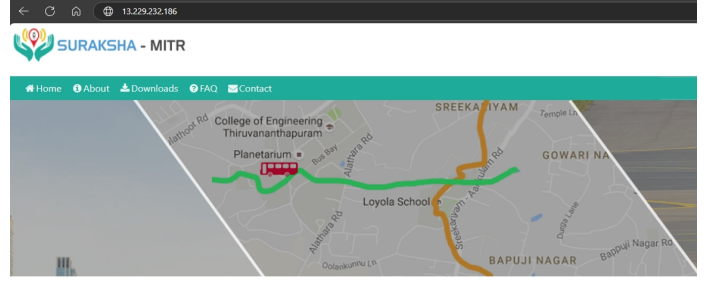


Fig. 8. LOC tracing result

C. Digital Forensics Investigation: Collecting Evidence

VTD device in the network is identified by its IP address, which makes inter-device communication possible. The IP address of the VTD device used here is 118.67.240.86, with port number 2021. The server IP is 13.229.232.186. If we use the server IP as a URL, we can find the Domain Name System (DNS) of the service provider. To find the eSIM service provider, looking into the Access Point Name (APN) session might help. Here, the eSIM service provider is Sensorise M2M. The reverse DNS lookup result is shown in Figure 9.



SURAKSHA - MITR

"Suraksha Mitr" developed by C-DAC is a highly scalable, comprehensive tracking solution for public transport vehicles designed for enhancing safety and enforcement. It is compatible with the AIS-140 standards published by MoRTH

C-DAC's proposal to MoRTH is for rollout "Suraksha Mitr" in all States/UTs with required customizations and can be deployed in any public cloud or State Data Centers. C-DAC has implemented "Suraksha Mitr"- GPS based vehicle tracking and monitoring system for ensuring safety and enforcement in public transport vehicles in Kerala.

Fig. 9. Reverse DNS lookup result

VIII. NOVEL ESIM FORENSICS FRAMEWORK

We can apply this framework to various M2M technologies utilizing eSIM, including VTD, OBD, smart metering systems, etc. Our forensic framework performs a comprehensive forensic investigation process in eSIM-connected IoT and M2M devices. The steps involved are shown in Figure 10. While developing this framework, we needed to factor in the critical nature, time sensitivity, and the security and privacy of the data being extracted.

IX. CHALLENGES AND CONSIDERATIONS

The challenges associated with this research include the potential risk of direct interaction with the eSIM, which could result in damage to the eUICC chip. Moreover, improper use of AT commands could compromise the eSIM profile. This section outlines the various challenges encountered during the initial setup stage and clearly describes the challenges faced during the evidence collection stages.

A. Software Challenges

Identifying suitable software tools for this task can be challenging, as most of the currently available tools are designed primarily for traditional SIM forensics.

B. Hardware Challenges

The eSIM integration in each M2M and IoT module varies based on hardware design. Another challenge is USB-to-serial data conversion, where data pin selection and pin mapping vary based on the type of chipset used in the hardware.

C. Encryption Challenges

eSIM technology is based on advanced encryption methods, including Advanced Encryption Standard (AES), Rivest-Shamir-Adleman (RSA) and Elliptic Curve Cryptography (ECC), to ensure secure communication and data protection [3]. These encryption techniques, coupled with 3GPP standards and the integration of Secure Elements (SE), establish

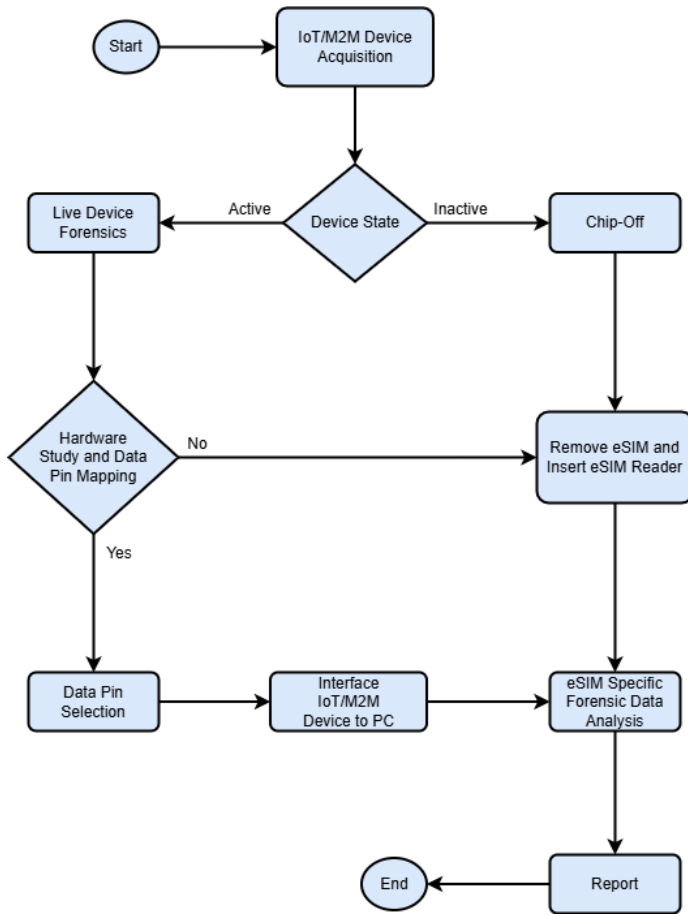


Fig. 10. Novel eSIM forensics framework for M2M and IoT

a robust security framework that safeguards sensitive data and prevents unauthorized access. Such measures make eSIM highly secure for handling sensitive information and interactions with Mobile Network Operators (MNOs).

However, despite this strong security framework, direct interaction with the eSIM profile, whether for updates, modifications, or profile switching, poses certain risks. Improper handling or unauthorized access can inadvertently damage the eSIM profile, resulting in a service disruption or loss of functionality. This concern becomes particularly significant during the collection of digital evidence, where such interactions could potentially disrupt forensic investigations. To mitigate these risks, strict adherence to secure handling protocols is imperative to preserve the integrity of the eSIM and ensure the success of the evidence collection process.

D. Future Challenges

Rapid developments in IoT/M2M technology restrict the development of unique solutions. This makes finding experts in this specific domain more challenging.

The aforementioned challenges, particularly encryption-related obstacles, significantly restrict access to confidential information stored in eSIM memory locations. These restrictions pose critical limitations to performing forensic analysis

in IoT and M2M technologies. To address these issues, this research proposes a novel approach involving manual deep packet inspection for eSIM forensics. This method aims to overcome existing barriers while maintaining the integrity of forensic investigations. Additionally, future research is planned to explore advanced techniques and methodologies to enable secure and authorized access to restricted eSIM memory locations. Such efforts will enhance the efficiency and reliability of eSIM forensic processes.

X. CONCLUSION AND FUTURE WORK

Research in IoT forensics, specifically in eSIM-integrated IoT and M2M technologies is limited. As eSIM technologies advance, they introduce novel challenges in IoT forensics. Additionally, leveraging eSIMs in both consumer electronics and machine-to-machine (M2M) communications is being streamlined swiftly. We proposed an effective forensic solution for eSIM-integrated IoT/M2M devices and outlined the methodology for conducting forensic investigations in M2M technologies with real-time experiments. Our Novel eSIM Forensic Framework successfully carried out DFI procedures in integrated eSIM with the GSM/GPRS module in VTD as part of the case study. We also clarified the type of data that is being collected when the framework is used. Currently, our work shows significant potential for implementation in sectors where digital security is crucial, such as banking (for high-value asset transportation), logistics (ensuring secure transit of valuable commodities), and defense. Additionally, this framework model can also be applied to standalone eSIM modules as their adoption gets accelerated.

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